TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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The Simplex Algorithm

The Simplex Algorithm is a mathematical method used to solve linear programming problems. It was developed by George Dantzig in 1947 and has since become one of the most widely used optimization techniques in various fields, including operations research, economics, and engineering.

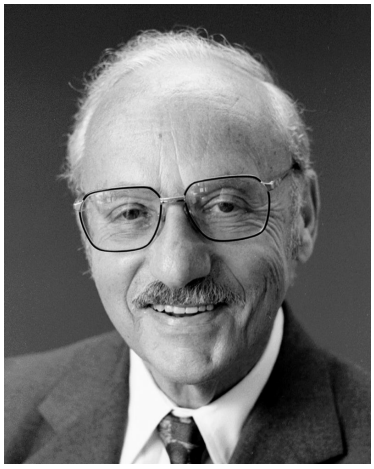


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Some name
- Amount of decision variables: 4
- Amount of restrictions: 3
- Total Amount of variables: 7

Equations

$$Max\ Z = 72.000000x_1 + 78.000000x_2 + -55.000000x_3$$
$$such\ that:$$
$$15.000000x_1 + -67.000000x_2 + -45.000000x_3 \geq 4700.000000$$
$$2.000000x_1 + -29.000000x_2 + -47.000000x_3 \geq 6100.000000$$

Initial Table

Initial Table

Z	x_1	x_2	x_3	s_4	s_5	s_6
1.0000	72.0000	78.0000	-55.0000	0.0000	0.0000	0.0000
0.0000	15.0000	-67.0000	-45.0000	0.0000	1.0000	4700.0000
0.0000	2.0000	-29.0000	-47.0000	0.0000	0.0000	6100.0000

Intermediate Tables

Final Table

Solution